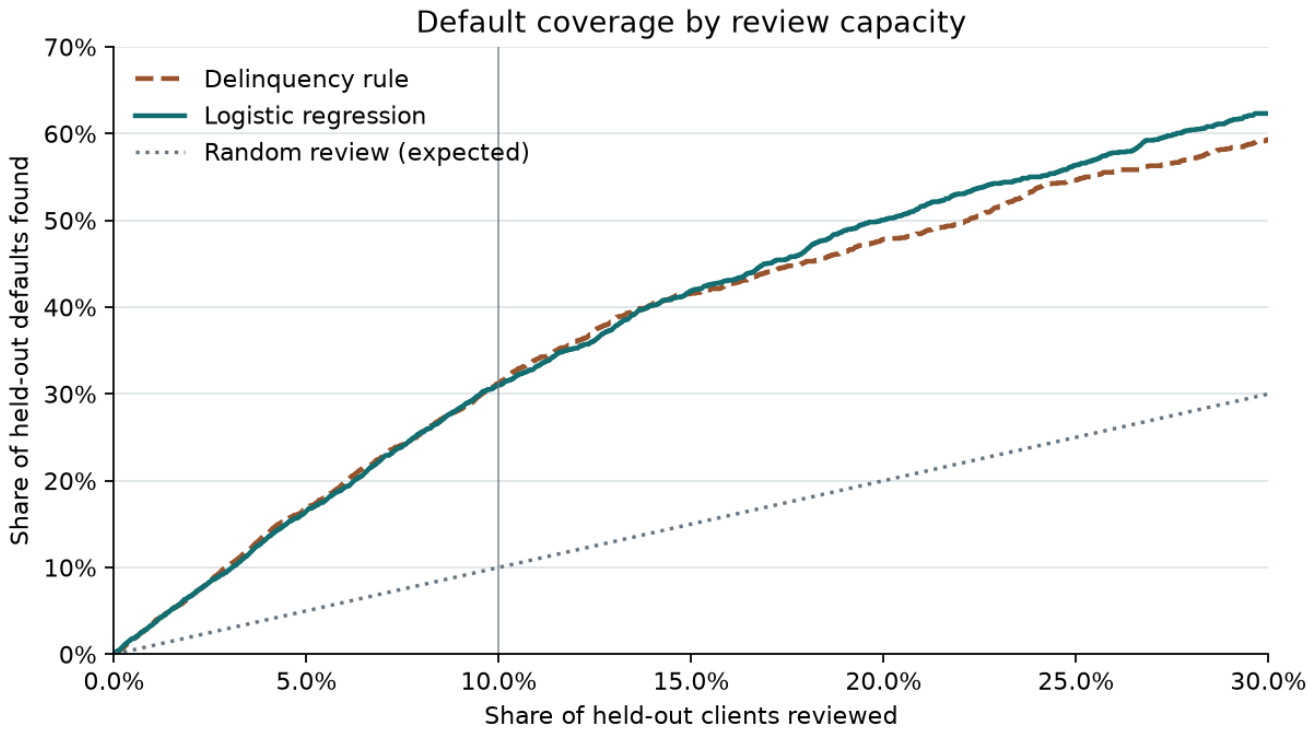
MODEL VALIDATION BRIEF

Credit card default early warning

Jiahui Zhang | Taiwan 2005 | Independent portfolio project

Keep the delinquency rule as the benchmark at 10% review capacity. On 6,000 held-out clients, the rule found 414 defaults and logistic regression found 412 across 600 reviews each. The paired 95% interval for the model's difference spans -17 to +10 defaults.

Queue	Found	Missed	Precision	Recall	ROC AUC
Delinquency rule	414	913	69.0%	31.2%	0.722
Logistic regression	412	915	68.7%	31.0%	0.766



Decision at the cutoff. The logistic model has higher ROC AUC on the held-out sample. Its top 600 accounts yield almost the same coverage. Compare queues at the actual review budget before changing the rule.

Validation design. Stratified 80/20 holdout. The model uses repayment, bill, payment and limit information available before the outcome month. Demographics and IDs are excluded. The interval resamples test clients and reselects both queues with the fitted scores.

Scope. One historical Taiwan cohort. A later lender cohort would be needed to verify performance, calibration and subgroup outcomes before any operational use.

Source | UCI Default of Credit Card Clients (Yeh, 2016)